COMPILER DESIGN

CSA1418

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QUESTION

S->CC

C-> cC |d

Construct the Parse Table of LALR Parser

CONSTRUCTION OF LALR(1) PARSING TABLE

Given Grammar:
S -> CC
C -> cC | d

STEP 1: AUGMENTED GRAMMAR

(0) S' -> S
(1) S -> CC
(2) C -> cC
(3) C -> d

STEP 2: FIRST SETS

FIRST(C) = {c, d}
FIRST(S) = {c, d}

STEP 3: CANONICAL LR(1) ITEM SETS

I0:
S' -> .S, $
S -> .CC, $
C -> .cC, c/d
C -> .d, c/d

GOTO(I0, S) = I1
GOTO(I0, C) = I2
GOTO(I0, c) = I3
GOTO(I0, d) = I4

I1:
S' -> S., $

I2:

S -> C.C, $
C -> .cC, $
C -> .d, $

GOTO(I2, C) = I5
GOTO(I2, c) = I6
GOTO(I2, d) = I7

I3:
C -> c.C, c/d
C -> .cC, c/d
C -> .d, c/d

GOTO(I3, C) = I8
GOTO(I3, c) = I3
GOTO(I3, d) = I4

I4:
C -> d., c/d

I5:
S -> CC., $

I6:
C -> c.C, $
C -> .cC, $
C -> .d, $

GOTO(I6, C) = I9
GOTO(I6, c) = I6
GOTO(I6, d) = I7

I7:
C -> d., $

I8:
C -> cC., c/d

I9:
C -> cC., $

Total canonical LR(1) states = 10.

STEP 4: MERGING LR(1) STATES

States with identical LR(0) cores are merged.

I3 and I6 are merged.
I4 and I7 are merged.
I8 and I9 are merged.

Merged states:

I36:
C -> c.C, c/d/$
C -> .cC, c/d/$
C -> .d, c/d/$

I47:
C -> d., c/d/$

I89:
C -> cC., c/d/$

Remaining states: I0, I1, I2, I5.

Total LALR(1) states = 7.

STEP 5: RENUMBERING STATES

I0 = 0
I1 = 1
I2 = 2
I3 + I6 = 3
I4 + I7 = 4
I5 = 5
I8 + I9 = 6

STEP 6: LALR(1) PARSING TABLE

State	c	d	$	S	C
0	s3	s4	-	1	2
1	-	-	acc	-	-
2	s3	s4	-	-	5
3	s3	s4	-	-	6
4	r3	r3	r3	-	-
5	-	-	r1	-	-
6	r2	r2	r2	-	-

Reduction rules:
r1: S -> CC
r2: C -> cC
r3: C -> d

s = Shift
r = Reduce
acc = Accept
- = Error

STEP 7: PARSING EXAMPLE

Input string: cdd$

STEP	STACK	INPUT	ACTION
1	0	cdd$	s3
2	0 c 3	dd$	s4
3	0 c 3 d 4	d$	r3
4	0 c 3 C 6	d$	r2
5	0 C 2	d$	s4
6	0 C 2 d 4	$	r3
7	0 C 2 C 5	$	r1
8	0 S 1	$	acc

The string cdd is accepted.

STEP 8: CONCLUSION

The given grammar has 10 canonical LR(1) states.
After merging identical LR(0) cores, 7 LALR(1) states remain.
The parsing table contains no shift-reduce or reduce-reduce conflicts.

Hence, the given grammar is LALR(1).